

Vaishnav Sreekumar

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AI Software Engineer — LLMs, AI Agents & Backend Systems

SUMMARY

Computer Science undergraduate graduating in 2027 with hands-on experience building AI agents, LLM-powered applications, backend APIs, cloud systems, and distributed data pipelines. Experienced with agent orchestration, tool execution, structured outputs, guardrails, REST APIs, Python, Java, AWS, and rapid experimentation with AI technologies. Strong debugging and systems background through OS development and open-source contributions, with a focus on turning real-world problems into working software.

EDUCATION

Vellore Institute of Technology (VIT), Bhopal

2023 – 2027

B.Tech

CGPA: 8.19/10

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming

TECHNICAL SKILLS

AI / LLM: LLMs, AI Agents, Agentic Workflows, Agent Orchestration, Prompt Engineering, Guardrails, Tool Use, Structured Outputs, Gemini API, OpenAI API, LangGraph4j

Languages: C, C++, Python, Go, Java, SQL, JavaScript/TypeScript

Backend & APIs: FastAPI, Spring Boot, REST APIs, PostgreSQL

Cloud & Infrastructure: AWS, Docker, Kubernetes, Terraform, Git, GitHub Actions, CI/CD, CloudWatch, Lambda, EventBridge

Data & Systems: Kafka, Elasticsearch, ELK Stack, Prometheus, Grafana, Operating Systems, Concurrency, Linux, Distributed Systems

Engineering: Testing, Debugging, Code Review, RBAC, Git

PROJECTS

AGENT FORGE – Multi-Agent Workflow System

Spring Boot, React Flow, LangGraph4j, Gemini

- Built a visual workflow system for graph-based AI agent orchestration, tool execution, and LLM task execution using LangGraph4j and a Spring Boot backend.
- Designed modular agent workflows that connect LLM-powered tasks with application tools through structured execution flows and backend APIs.

SentinelFinOps – AI-Powered Cloud Governance Platform

Python, FastAPI, OpenAI API, AWS, Terraform

- Built an AI-powered cloud governance platform combining AWS telemetry with LLM reasoning to automate cloud cost and operational analysis.
- Developed an AI Gateway with provider abstraction, schema validation, guardrails, and structured outputs for reliable LLM interactions.
- Implemented multi-region and cross-account scanning, IAM-based access, audit logging, health checks, automated workflows, and GitHub Actions CI/CD with 116+ unit and integration tests.

SentinelX – Multi-Tenant SIEM & Observability Platform

Python, Kafka, ELK, Prometheus, Grafana

- Built a real-time event pipeline for ingesting, processing, indexing, and monitoring high-volume system and security telemetry using Kafka.
- Implemented multi-tenant isolation, Elasticsearch-backed search and analysis, and Grafana dashboards for monitoring and visualization.

ZeroOS – Experimental Operating System Kernel

C, C++, x86, OS, Concurrency

- Developed an experimental operating system kernel implementing kernel-level threading, scheduling, synchronization, interrupt handling, and timer facilities.
- Implemented cooperative and LAPIC-based preemptive scheduling with WaitQueues, mutexes, condition variables, and timer-based sleep, achieving 82/82 passing tests.

OPEN SOURCE

Kubernetes SIGs – Upstream Contributor

- Authored and merged an upstream PR implementing granular RBAC status authorization for node-local drivers, addressing authorization issues for Kubernetes v1.36.
- Worked through upstream code review, CI checks, testing, and Kubernetes contribution standards.

Kerno – Open Source Contributor

- Implemented kerno doctor `--list-rules` with automated test coverage and contributed through code review and CI workflows.

CERTIFICATION

AWS Certified Cloud Practitioner